

Document 1: Canvas & Core Data Model Specification

1.1 Fixed Analysis Grid Mechanics

- **Grid Resolution:** 128×128 cells
- **Total Cell Count:** 16,384 cells
- **Memory Alignment:** Power-of-Two (2^7)
- **Data Type:** 8-bit Unsigned Integer (`uint8_t`)
- **Memory Allocation:** Exactly 16 KB per grid instance

1.2 Viewport Coordinate Mapping

The GUI window serves as a viewport transforming screen coordinates to the static internal 128×128 coordinate space.

Let W_{canvas} and H_{canvas} represent the physical dimensions of the UI window. Let (X_{ui}, Y_{ui}) represent the raw input coordinate. The target analysis grid coordinate (X_{grid}, Y_{grid}) is computed via:

$$X_{grid} = \lfloor (X_{ui} \times 128) / W_{canvas} \rfloor$$

$$Y_{grid} = \lfloor (Y_{ui} \times 128) / H_{canvas} \rfloor$$

1.3 Input Rasterization Pipeline

User mouse or stylus strokes are captured as sequential vector paths and rasterized into the 16 KB grid using an integer line interpolation algorithm. Data writes occur immediately upon coordinate contact.